IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2017

EEE PART II: MEng, BEng and ACGI

MATHEMATICS 2B (E-STREAM AND I-STREAM)

Corrected copy

Friday, 26 May 2:00 pm

Time allowed: 1:30 hours

There are TWO questions on this paper.

Answer TWO questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : B. Clerckx
Second Marker(s) : D. Nucinkis

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THE QUESTIONS

[25]

1. Consider two continuous random variables X and Y characterized by the following joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} 3(x+y), & 0<x<1,\ 0<y<1,\ 0<x+y<1, \\ 0, & \textit{otherwise.} \end{cases}$$

a) Compute the probability $P(X+Y \leq 0.5)$.

[4]

b) Compute the marginal probability density function of X.

[4]

c) Compute the expectation of X, i.e. $E(X)$, and the variance of X, i.e. $\mathrm{Var}(X)$.

[4]

d) Compute the marginal probability density function of Y.

[2]

e) Compute the expectation of Y, i.e. $E(Y)$, and the variance of Y, i.e. $\mathrm{Var}(Y)$.

[2]

f) Compute the covariance between X and Y, i.e. $\mathrm{Cov}(X,Y)$, and the correlation coefficient between X and Y, i.e. $\mathrm{Corr}(X,Y)$.

[2]

g) Are X and Y uncorrelated? Independent? Provide your reasoning.

[2]

h) Compute the conditional probability density function of Y given $X = x$.

[2]

i) Compute the conditional expectation of Y given $X = x$.

[3]

[25]

2. a) Consider the independent and identically distributed (i.i.d.) random variables $X_1, X_2, \ldots, X_n$ characterized by their probability density function

$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & otherwise. \end{cases}$$

i) Using the Central Limit Theorem (CLT), compute an approximate value of $P\left(\prod_{i=1}^{n} X_i \leq \exp\left(-\frac{n}{2} + 0.5\sqrt{n}\right)\right)$. Provide your reasoning.

[7]

ii) Compute the probability density function of the random variable $U = X_1 X_2$. Provide your reasoning.

[6]

iii) Compute the expectation of U, i.e. $E(U)$. Provide your reasoning.

[2]

b) i) Assume $X \sim N(0,1)$ and take $Y = X^4$. Are X and Y uncorrelated? Independent? Provide your reasoning. [5]

ii) Consider the following statement: If X is a continuous random variable with first moment m_1 and second moment m_2, then we have $m_1^2 = m_2$. Is the statement correct? If yes, provide a proof. If not, correct the statement and provide a proof. [5]

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